

MISSION REPORT

Name: Andrea Betterle & Jesús Casado Rodríguez

Directorate E - Unit E.1 Disaster Risk Management

Objectives of mission: Participation and oral presentation at the European Geophysical Union annual assembly 2023

Places, dates and organisations visited: Austria Centre, Vienna, Austria (23-27 April 2023)

People met and talked to:

Andrea Betterle:

Jose Salinas (RMS), Paul Bates (University of Bristol), Basudev Bisval (Indian institute of Technology, Bombay), Bernard Bauer Marschallinger (TU Wien), Florian Roth (TU Wien), Christian Krullikowski (DLR), Wouter Berghuijs (Free university of Amsterdam), Diego Avesani (University of Trento), Paolo Salandin (University of Padova), Gianluca Botter (University of Padova), Enrico Bertuzzo (University of Venice), Sebastian Gnann (University of Potsdam).

Jesús Casado Rodríguez:

Gwyneth Mathews (ECMWF), Ilias Pechlivanidis (SMHI), Michaela Mikulickova (SHMU), Jan, Bliefernicht (Universität Augsburg), Kor de Jong (Utrecht University), Oldrich Rakovec (UFZ), Pau Wiersma (Université de Lausanne), Sarah Hanus (UZH), Frederik Kratzert (Google), Wouter Knoben (University of Saskatchewan), Cristina Prieto Sierra (Universidad de Cantabria), Larisa Tarasova (UFZ), Hannes Müller-Thomy (TU Braunschweig), Laurène Bouaziz (Deltares), Edgar Espitia Sarmiento (Universität Giessen).

Other participants:

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Main results of the mission:

Andrea Betterle:

As the largest European conference on earth sciences, EGU is the ideal stage to present and discuss scientific topics related to geosciences and to foster collaborations across institutions and disciplines.

My participation to EGU had two main objectives:

1. Present the results of the work I'm currently performing on generating flood extent maps featuring water depth at a global scale based on flood delineations obtained from the recently released Global Flood Monitoring (GFM) system of the Copernicus EMS. Specifically, the presentation was entitled "A parsimonious approach to estimate flood depths —

also in urban areas — for satellite-based flood maps” which took place in the session “Remote sensing for flood dynamics monitoring and flood mapping”. Further details and the abstract are available at <https://meetingorganizer.copernicus.org/EGU23/EGU23-1603.html>

2. Discussions and networking with expert in the field of flood mapping based on SAR data as well as with flood hazard modellers. Additionally I met in person for the first time some of the members of the GFM developing team (specifically: Bernard Bauer Marschallinger (TU Wien), Florian Roth (TU Wien), Christian Krullikowski (DLR)).

Jesús Casado Rodríguez

My participation to EGU was two-folded:

1. I gave an oral presentation in the session *HS4.3 Probabilistic hydro-meteorological forecasts*. I presented the analysis and optimization of the formal flood notification criteria used to warn partners of the European Flood Awareness System (EFAS).
2. I presented a poster in the session *HS2.5.1 Large-scale hydrology*. The poster showed the latest developments of the Global Flood Awareness System (GloFAS v4.0) in which the team has worked during the last couple of years, including the creation of a new set of static maps, the developments in the hydrological model LISFLOOD-OS, and the calibration, validation and regionalization of model parameters throughout the globe.

Additional information:

Andrea Betterle:

Specific discussion during the week included:

Day 1: Discussion with Jose Salinas (RMS) on flood impact assessment and the practices adopted in flood risk modelling at RMS

Day 2: Update on some issues of GFM with some members of the GFM developing team (i.e. Bernard Bauer Marschallinger (TU Wien), Florian Roth (TU Wien), Christian Krullikowski (DLR))

Day 3: Discussion with Prof. Basudev Bisval (Technical university of Bombay) on possible ways to combine GFM-based water depth estimates with the current approach used in India to delineate floods.

Day 4: I delivered my oral presentation which was followed by some discussions with participants in the session.

Day 5: Discussion with Paul Bates (University of Bristol) on some technical aspects related to the estimation of water depth in flooded areas using DTM and flood delineation areas.

Jesús Casado Rodríguez:

- Day 1. I delivered the oral presentation and had feedback afterwards from various attendants. On this day there were several sessions related to the use of machine learning and deep learning in hydrology (*HS3.1 Hydroinformatics*, *HS3.3 Deep Learning in Hydrology*), which are of interest for future developments of LISFLOOD-OS.
- Day 2. I presented the poster, which attracted notorious attention, in particular the availability of the new static maps. Among others, I attended the oral presentation on session *HS2.5.1 Large-scale hydrology*, closely related to the work presented in the poster.
- Day 3. I attended a series of sessions related to the GREAT project (Green Deal Data Space), in which I will be involved in the following weeks. I contacted Kor de Jong, leader of the hydrological part of the project and main developer of the LUE modelling framework, which could be of interest for the future development of LISFLOOD-OS.
- Day 4. Among others, I attended session *HS2.1.8 Large-sample hydrology*, which is of special interest for GloFAS and the possible collaboration with the project Caravan.
- Day 5. In session *HS2.2.1 Advancing process representation for hydrological modelling* I met Lauréne Bouaziz (Deltares), who presented a comparison between several routing methods (local inertia 1D and local inertia 1D2D) to replace kinematic wave from the model Wflow, a similar idea to one of the ongoing developments in LISFLOOD-OS. In session *HS4.4 I Operational forecasting and warning systems for natural hazards and climate emergency* I listened and talked to people working on other warning systems similar to EFAS and GloFAS.

For mission outside of the Community

Contact with the Delegation of the Commission (in Washington, Tokyo, etc.) has been taken up ? ☐ yes ☐ no